

JX1 humanoid project · research paper · 6 October 2026

Building humanoid joint actuators in-house

How the motors work, how much torque a humanoid needs, how to design, wind and build your own RobStride-class actuators in India, and what it costs compared with buying from China

3 classes

JXA-40 · JXA-120 · JXA-360: 40 / 120 / 360 N·m peak, covering JX1 and the knee of a 70 kg full-size humanoid

±3-20 %

our calculated JXA-120 vs the RobStride RS04 datasheet (K_t , K_e , R, L, peak current)

₹30,292

JXA-120 in-house batch cost, vs ₹24,091 for an RS04 landed via a China distributor

₹3.2 lakh

garage kit to wind, assemble and test prototypes (no factory needed)

≈ 875 / yr

volume at which an Indian line beats the landed import price

₹11,357

estimated cost for RobStride to make one RS04 (range ₹8,923–₹20,280)

Abstract. We explain from first principles how a permanent-magnet motor turns current into torque, survey the actuators used by today's humanoids (RobStride, Damiao, Unitree, CubeMars, MIT, Tesla), and derive torque needs from JX1's own analysis, 29 robots and human biomechanics. A three-class family of outer-rotor, single-stage planetary actuators (JXA-40/120/360) is designed with an analytic model that reproduces the RS04 datasheet within 3-20 %. We give a winding procedure a technician can follow, a build sequence that uses Indian job shops instead of a factory, a test plan, and a cost model at prototype, batch and 1,000-per-year scale, compared with buying and with Chinese makers' estimated costs. Conclusion: building is feasible and valuable for learning, the 360 N·m class and supply security. It is not cheaper than RobStride below ≈ 1,000 units a year, and Chinese makers' advantage is volume and supply chain, not wages.

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1. Goal and scope

JX1 spends 77 % of its parts budget on actuators bought from RobStride ([bom/cost_summary.md](#)). This paper asks whether those actuators, and the bigger ones a full-size humanoid needs, can be designed and built in-house in India with little tooling, how to do it, and what it costs compared with buying and with how Chinese makers do it. It covers:

- the physics (magnetism, Lorentz force, induction, torque and speed limits): Section 2;
- today's actuators and what is inside them: Section 3;
- torque needs from JX1 up to 70 kg humanoids: Section 4;
- a complete first-pass design of three actuator classes, with winding data: Sections 5-6;
- how to make each part with job shops and a small workshop: Section 7;
- costs at three scales vs RobStride and vs Chinese factory costs: Section 8; then testing, risks and a phased plan.

Labels follow the repository convention: VERIFIED, CALCULATED, ESTIMATED, UNVERIFIED. Nothing here has been built yet; every design number is a calculation to be confirmed on the bench.

2. How an electric motor makes torque (from magnetism up)

2.1 Four facts of electromagnetism you need

1. **A current makes a magnetic field.** Wind wire around an iron tooth and push current through it: the tooth becomes an electromagnet. Its strength is set by the *ampere-turns* $N \cdot I$ (turns $\times$ current), not by the current alone. Iron guides the field about 1000 times more easily than air, which is why motors are mostly iron with a very thin air gap.
2. **Opposite poles attract, like poles repel.** A permanent magnet (NdFeB, "neodymium") is a magnetic field you get for free, with no current and no heat. NdFeB grade N42 has a remanence $B_r \approx 1.3$ tesla.
3. **Force on a current in a field (Lorentz):** $F = B \cdot I \cdot \ell$. A wire of length ℓ carrying current I in flux density B feels a sideways force. Every torque in this paper comes from this one line.
4. **A changing field makes a voltage (Faraday):** $e = -d\lambda/dt$. When the magnets sweep past the coils, they induce a voltage, the *back-EMF*. It grows with speed and is what limits how fast the motor can spin on a 48 V battery. The same physics run backwards is a generator: turn a motor by hand and it makes electricity.

2.2 From force to torque: the one equation that sizes a motor

Add up the Lorentz force on every conductor in the air gap and multiply by the radius. The result depends on the air-gap **shear stress** σ (newtons of tangential pull per square metre of rotor surface) and the rotor size:

$$T = \sigma \cdot (\pi \cdot D \cdot L) \cdot D/2 = (\pi/2) \cdot \sigma \cdot D^2 \cdot L$$

D is the air-gap diameter and L the stack length (the length of the iron). $\sigma = (\text{magnetic loading } B) \times (\text{electric loading } A, \text{ amperes of current per metre of circumference}) / 2$. Air-cooled servo motors run $\sigma \approx 5\text{--}15$ kPa continuously and 30–60 kPa for a few seconds at peak. Our three designs reach 42 / 46 / 45 kPa at peak. **Torque grows with D^2 , but only with L** , which is why robot actuators are "pancakes": large diameter, short length.

2.3 Brushless motor = magnets on the rotor, copper on the stator

A brushless DC (BLDC) or permanent-magnet synchronous motor (PMSM; same hardware) has:

- a **stator**: a stack of thin, insulated silicon-steel sheets (laminations, 0.2–0.35 mm) with *teeth*; copper coils are wound around the teeth;
- a **rotor**: a steel ring carrying 2p magnets of alternating polarity (p = pole pairs);
- three **phases** (A, B, C) of coils, driven by a three-phase inverter (six MOSFETs) at a frequency $f = p \cdot \text{rpm} / 60$.

The controller always energises the coils so that the stator field sits 90 electrical degrees ahead of the rotor field. The rotor keeps chasing it, and that steady pull is the torque. This is **field-oriented control (FOC)**. It needs the rotor angle at all times, so every actuator carries a magnetic angle encoder.

Robot actuators use the **outer-rotor** ("outrunner") layout: stator inside, magnet ring outside, as in drone motors. For a given outside diameter it gives the largest air-gap diameter D and so the most torque ($T \propto D^2$). It is also the easiest stator to wind by hand, because the teeth point outwards and you can reach every slot.

2.4 Slots, poles and the winding factor

Robot motors use **tooth-concentrated windings**: each coil is wound around one tooth. They have short end turns (less copper, less heat), they are easy to wind, and they work with slot/pole combinations close to each other, such as 12N14P, 24N28P, 36N40P and 48N44P (N = slots/teeth, P = magnet poles). The **winding factor** k_w (0–1) says how much of the copper's ampere-turns produce useful torque. Our script finds the coil-to-phase assignment by the "star of slots" method:

Class	Slots / poles	k_w	Repeating units	Cogging order (LCM)	Why
JXA-40	24N28P	0.933	2	168	same family as 12N14P drone motors; RS06/CyberGear are 28-pole
JXA-120	36N40P	0.945	4	360	highest k_w of the 36-slot options, 4 symmetric units allow 4 parallel groups
JXA-360	48N44P	0.949	2	528	many poles for a large diameter; balanced, smooth (LCM 528)

A high LCM of slots and poles means many small cogging steps that mostly cancel, giving smooth torque. An even number of repeating units means the magnetic pulls balance and the rotor is not pulled to one side (no unbalanced magnetic pull).

2.5 The motor constants that matter for a robot

$$T = K_t \cdot I \quad e = K_e \cdot \omega \quad (K_t = K_e \text{ in SI units}) \quad P_{\text{copper}} = 3 \cdot I_{\text{rms}}^2 \cdot R \quad K_m = T / \sqrt{P_{\text{copper}}}$$

- **K_t** (N·m per ampere): more turns give more torque per amp, but also more back-EMF and so a lower top speed on 48 V. The number of turns is therefore chosen *from the speed you need*, and the wire is then made as thick as the slot allows.
- **K_m** (motor constant, N·m/√W): torque per square root of heat. It depends only on how much copper and magnet you put in the gap, *not on the number of turns*. It is the true figure of merit: heat = $(T/K_m)^2$. Doubling the torque costs four times the heat. K_m grows roughly with $D^{2.5} \cdot L$, so a big, flat motor wins.
- **Thermal limit**: the class-H enamel survives 180–200 °C, NdFeB SH grade 150 °C. We cap the winding at 120 °C. Continuous torque is whatever torque keeps the copper below that with the heat the actuator can shed (here 2.00 / 1.05 / 0.62 K/W from winding to air when bolted to an aluminium limb).

2.6 Why a low gear ratio ("quasi-direct drive")

Legged robots hit the ground hard and need to feel forces through the motor current. A gear ratio G multiplies torque by G, but it also multiplies the rotor's inertia by G^2 and the friction by roughly G. With G

≈ 6–10 (one planetary stage) the joint stays **back-drivable**: you can push it by hand, impacts do not strip teeth, and current is a good measure of joint torque. That is why MIT Cheetah, Unitree, RobStride, Damiao and Berkeley Humanoid all use 6–10:1 single-stage planetaries for legs. We use 9:1 for the 40 and 120 N·m classes and 10:1 for the 360 N·m class. The ratio of a planetary with the ring gear fixed is

$$G = 1 + Z_{\text{ring}} / Z_{\text{sun}} \quad \text{with} \quad Z_{\text{ring}} = Z_{\text{sun}} + 2 \cdot Z_{\text{planet}} \quad \text{and} \quad (Z_{\text{sun}} + Z_{\text{ring}}) / N_{\text{planets}} = \text{whole number}$$

2.7 The 48 V speed limit and the torque-speed curve

The inverter can put at most about $V_{\text{dc}}/\sqrt{3} \approx 26 \text{ V}$ (peak, per phase) across a winding. As speed rises, back-EMF uses up that voltage until no current, and so no torque, can be pushed in: that is the no-load speed. Below it, the current limit sets the flat "peak torque" line (Fig. 2). Field weakening (negative d-axis current) can stretch the top speed by 20–40 % at the cost of heat. The same trade-off explains every datasheet: RS04 (120 N·m, 21 rad/s) and RS06 (36 N·m, 50 rad/s) use different turns on similar hardware.

3. What is on the market, and what is inside it

The existing repository survey ([research/raw/actuator_technology_raw.md](#), ≈110 actuators) and two new logs ([actuator_internals_and_torque_raw.md](#), [china_actuator_manufacturing_economics_raw.md](#)) were condensed into Table 1. Almost every walking robot today uses the same recipe: an outer-rotor or frameless PM motor, a 6–10:1 planetary (two stages, 14–25:1, for the biggest joints), a crossed-roller output bearing, two magnetic encoders and a FOC driver on the back, talking CAN or RS-485.

Actuator	Peak N·m	Rated N·m	No-load rad/s	Mass g	Reducer	Price	Notes
RobStride RS00 / RS02 / RS06	14 / 17 / 36	5 / 6 / 11	33 / 43 / 50	310 / 405 / 621	10 / 7.75 / 9 planetary	\$125 / \$145 / \$210 (¥598 / ¥699 / ¥849)	Beijing; ex-Xiaomi CyberGear team
RobStride RS03 / RS04	60 / 120	20 / 40	20.4 / 20.9	880 / 1,420	9:1 planetary, 42 poles	\$225 / \$255 (¥999 / ¥1,199)	largest RobStride model (no >120 N·m unit)
Damiao DM-J10010L / J10422P	120 / 400	40 / 100	20.9 / 12.6	1,372 / 2,700	10:1 / 22:1	\$265 / \$480 (reseller)	Shenzhen; CAN-FD on new models
Unitree GO-M8010-6 / IM6014	23.7 / 34.4	–	30 / 54	530 / 535	6.33 / 12.66	\$369 / \$269	RS-485; Go2: outer rotor 36N42P, 6.22:1
Unitree G1 knee / H1 knee	90–139 / 360	–	~20 / 14	–	2-stage 22.5:1 / –	not sold separately	G1: 18N16P inner rotor (teardown, unverified)
CubeMars AK80-9 / AK10-9	18 / 48	9 / 18	– / 10.5	485 / 960	9:1 planetary, 36N42P Ø98	₹79,739 / ₹1,04,809 (Robu)	3–8× RobStride per N·m in India
MIT Humanoid (U10 / U12 module)	33.6 / 68	–	55 / 45	619 / 1,174	6:1 planetary (+ belts)	research	knee 136 N·m via 12:1 incl. belt
Berkeley Humanoid Lite 6512	~20–28	–	–	–	15:1 printed cycloid	US\$157–188	open source; walks at 6 N·m caps
Tesla Optimus rotary (3 sizes)	20 / 110 / 180	–	–	–	strain-wave + clutch + crossed roller	in-house	plus linear 500 / 3,900 / 8,000 N roller screws

Table 1. Representative actuators (VERIFIED datasheets/official sites unless the notes say otherwise; full sources in the raw logs).

3.1 What we know about the inside of a RobStride RS04

No teardown of a RobStride actuator has been published (English or Chinese searches, October 2026). RobStride's own website code gives 42 poles, 9:1, "machined steel" gears and the electrical constants. Every 42-pole motor of this size with a published slot count (CubeMars AK10-9, Unitree Go2, T-Motor U10) is 36N42P, so RS04 is very probably the same. We used the RS04 numbers to check our own motor model (Table 2): the calculated JXA-120 lands within 3–20 % of the RS04 datasheet on every electrical constant, which is about as close as an analytic model gets. This is the main evidence that the designs in Section 5 are realistic.

Quantity (RS04 datasheet → our JXA-120 model)	RobStride RS04 (published)	JXA-120 (calculated)	Comment
Poles / slots	42 / 36 (slots inferred)	40 / 36	bought-core variant uses 36N42P
Gear ratio	9	9 (12 / 42 / 96 teeth)	same
Kt at output, N·m/A _{rms}	2.10	2.31	+10 %
Back-EMF, V _{rms} line-line per 1000 motor rpm	16.9	16.3	−3 %
Phase resistance	≈ 80 mΩ (0.16 Ω line-line)	48 mΩ at 20 °C	ours has a 20 mm stack: more copper
Phase inductance	≈ 105 μH (0.211 mH line-line)	85 μH	−19 %
Peak phase current	90 A	86 A	−4 %
No-load speed at 48 V	20.9 rad/s	23.1 rad/s	+11 %
Mass	1.42 kg	1.91 kg	first in-house design is heavier

Table 2. Model check against the RS04 datasheet (RobStride site code and manual 260713).

4. How much torque does a humanoid need?

Torque needs scale with mass × leg length × how violently the robot moves. Three sources were combined: JX1's own rigid-body analysis (walking up to 0.79 m/s, squats, single-leg stands, with a 1.5× dynamic margin; [calculations/results/iter1_B_knee_and_pitch_RS04/](#)), the published limits of 29 commercial and open humanoids, and human biomechanics.

4.1 JX1 (1.2 m, 27.5 kg)

JX1 joint (27.5 kg robot)	Required peak / continuous / speed	Today (bought)	In-house replacement
Hip pitch	85 N·m / 26 N·m / 12 rad/s	RS04 120 N·m	JXA-120
Knee	71 / 30 / 12	RS04 120	JXA-120
Hip roll	50 / 17 / 6	RS03 60	JXA-120 (one class instead of two)
Hip yaw	37 / 8 / 6	RS06 36 (3 % short)	JXA-40 (41 N·m: closes open issue OI-2)
Ankle motors (×2 per ankle)	32 / 3.4 / 8.8	RS06 36	JXA-40
Waist yaw	≈ 30	RS06 36	JXA-40
Shoulders, elbows	≤ 17	RS02 / RS00	keep buying (small, cheap per unit)

Table 3. JX1 requirements (CALCULATED, [actuators/actuator_selection.md](#)) and the in-house class that covers each joint.

4.2 The biggest humanoids (1.6–1.9 m, 50–70 kg)

Robot	Mass	Knee	Hip pitch	Hip roll / yaw	Ankle	Knee speed
Unitree H1 / H1-2	47 / 70 kg	360 N·m	220	220	59 / 2 × 75	14 rad/s
Fourier GR-2 / GR-3	63 / 70 kg	366	366	95–140	54–59	6.5 rad/s
UBTech Walker S2	70 kg	225 peak / 75 rated	225	225 / 65	2 × 65	8.4 rad/s
AgiBot A2 Lite	64 kg	270	–	–	–	–
Figure 02 (unverified)	≈ 70 kg	150	150	–	–	–
Fourier GR-1T2	≈ 53 kg	135	135	84 / 65	42	18.5 rad/s
Tesla Optimus (unverified classes)	≈ 57–73 kg	linear 8,000 N actuator via four-bar	rotary 180 N·m	110 N·m	linear 3,900 N	–

Table 4. Full-size humanoid joint limits (VERIFIED from official model files/SDKs unless marked; actuator_internals_and_torque_raw.md §2).

Human, per kg body mass (75 kg adult)	Hip	Knee	Ankle
Walking	0.5–1.0 N·m/kg	0.46–1.27	1.2–1.9
Stairs	–	≈ 1.0	1.4–1.8
Rising from a chair	0.24–1.92	0.51–1.97	–
Running 3.5 m/s	2.0	3.1	2.9
Sprinting ≈ 9 m/s	4.1	3.6	4.0

Table 5. Human joint moments (Schache 2011, Yoshioka 2014, arXiv 2511.06796). A 70 kg robot that should run, jump or get up from the floor needs about 3–5 N·m/kg at the knee, i.e. 200–360 N·m.

4.3 Three classes cover everything

Peak knee torque divided by robot mass ranges from 2.1 (Walker S2) to 7.7 N·m/kg (H1) across the field. We therefore define three classes. Together they cover JX1 completely and every leg joint of the largest humanoids, including the H1-class knee:

- **JXA-40:** 40 N·m peak, ≥ 40 rad/s. RS06 class. JX1 ankle, hip yaw, waist; arms of a full-size robot.
- **JXA-120:** 120 N·m peak, ≥ 20 rad/s. RS04 class. JX1 hip pitch, knee, hip roll; hip roll/yaw, ankle, shoulder, waist of a full-size robot.
- **JXA-360:** 360 N·m peak, ≥ 15 rad/s. Unitree H1 knee / Fourier GR-2 class. Knee and hip pitch of a 1.7–1.8 m, 50–70 kg robot. RobStride does not make anything this big.

5. The JXA actuator family: a design you can build

The design script `actuators/inhouse/qdd_design.py` sizes each class from first principles (Section 2): magnet circuit → air-gap flux → flux linkage → K_t ; slot geometry → copper fill → resistance and K_m ; inductance → torque-speed at 48 V; a lumped thermal resistance → continuous torque; Lewis bending and Hertz contact → gear safety factors; part volumes → mass and the material bill used by the cost model. Turns are chosen to meet the speed target with 12 % headroom, and the wire is then made as thick as a realistic hand/needle-wound fill of 0.42 (bare copper / slot area) allows. Saturation at peak current is calibrated on RS04 (its K_t drops ≈ 10 % between rated and peak current). All values are **CALCULATED**; expect ± 15 -30 % until a prototype is measured.

	JXA-40	JXA-120	JXA-360
Role	ankle, hip yaw, waist, shoulder (1.2 m robot); arms (1.7 m robot)	hip pitch, knee (1.2 m robot); hip roll/yaw, ankle (1.7 m robot)	knee, hip pitch (1.7-1.8 m, 50-70 kg robot)
Benchmark	RobStride RS06 (36/11 N.m, 621 g, 88 mm)	RobStride RS04 (120/40 N.m, 1420 g, 120 mm)	Unitree H1 M107 knee (360 N.m)
Motor: slots / poles	24N28P, outer rotor	36N40P, outer rotor	48N44P, outer rotor
Stator OD × stack	72 × 14 mm	100 × 20 mm	140 × 28 mm
Rotor OD / stator bore	82.0 / 43.8 mm	111.1 / 63.7 mm	153.3 / 102.0 mm
Air gap / magnet thickness	0.5 / 2.5 mm N42SH	0.55 / 3.0 mm N42SH	0.6 / 3.5 mm N42SH
Tooth width / slot depth / yoke	5.3 / 11 / 2.2 mm	5.0 / 15 / 2.2 mm	5.3 / 15 / 2.8 mm
Air-gap flux density (80 °C)	0.94 T	0.95 T	0.96 T
Gear: sun / planet / ring × 3 planets	12 / 42 / 96, module 0.7, face 10 mm	12 / 42 / 96, module 1.0, face 13 mm	12 / 48 / 108, module 1.25, face 20 mm
Gear ratio	9 : 1	9 : 1	10 : 1
Peak torque (output)	41 N·m at 59 A peak	125 N·m at 86 A peak	371 N·m at 206 A peak
Continuous torque (120 °C winding)	13 N·m	46 N·m	147 N·m
No-load speed at 48 V	47.7 rad/s (456 rpm)	23.1 rad/s (220 rpm)	18.6 rad/s (177 rpm)
K_t at output	1.11 N·m/A _{rms}	2.31 N·m/A _{rms}	2.86 N·m/A _{rms}
K_m motor / at output	0.25 / 2.1 N·m/√W	0.62 / 5.3 N·m/√W	1.40 / 13.3 N·m/√W
R phase 20 °C / L phase	69 mΩ / 72 μH	48 mΩ / 85 μH	12 mΩ / 24 μH
Copper loss at peak / time at peak (adiabatic)	472 W / 1.5 s	703 W / 2.3 s	988 W / 3.6 s
Current density peak / continuous	95 / 19 A/mm ²	76 / 18 A/mm ²	61 / 15 A/mm ²
Gear safety: bending static at peak / fatigue at cont. / contact at cont.	3.4 / 4.6 / 1.89	3.0 / 3.5 / 1.64	2.7 / 2.9 / 1.52
Reflected rotor inertia at output	192 kg·cm ²	749 kg·cm ²	4173 kg·cm ²
Envelope Ø × length	90 × 58 mm	119 × 67 mm	161 × 83 mm
Mass (estimated)	0.88 kg (47 N·m/kg)	1.91 kg (66 N·m/kg)	4.27 kg (87 N·m/kg)

Table 6. JXA family, first-pass design (results/design.json).

JXA-120 section (to scale): outer-rotor motor + single-stage 9:1 planetary

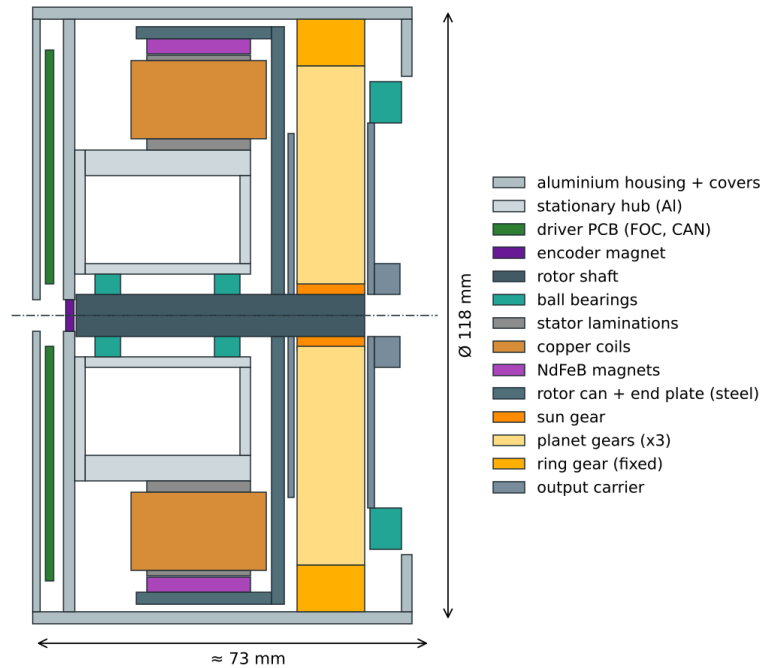


Figure 1. JXA-120 section, drawn to scale from the calculated dimensions. The stator is fixed on an aluminium hub; the outer magnet rotor drives the sun gear; the ring gear is part of the housing; the carrier is the output, held by one large bearing. The driver board and both encoders sit on the back cover.

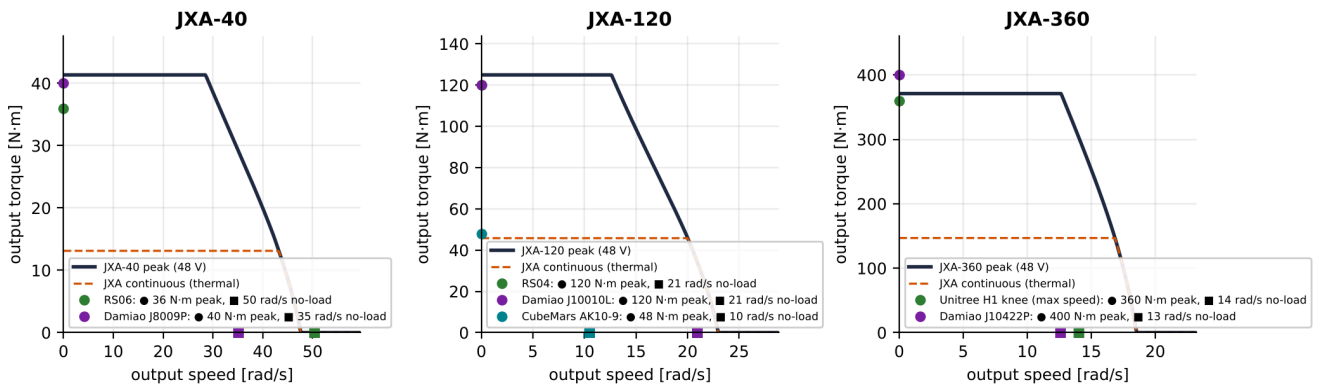


Figure 2. Calculated torque-speed envelopes at 48 V (field weakening off) and thermal continuous torque, with commercial benchmarks: ● peak torque, ■ no-load speed.

5.1 What the numbers say

- All three classes meet their peak, continuous and speed targets on paper. JXA-120's constants are within 3–20 % of RS04's datasheet (Table 2), and JXA-40's output K_t (1.11 N·m/A) matches RS06's (1.10).
- Our first design is heavier:** JXA-120 $\approx$ 1.9 kg vs 1.42 kg for RS04, mostly from the solid planets, thick housings and a longer stack. Lightening holes in the planets, a stepped-planet (compound) stage, 0.2 mm laminations and a die-cast housing are the second-iteration levers. A 34 % heavier knee actuator is acceptable on JX1 for a first build (the extra $\approx$ 0.5 kg per joint is $\approx$ 2 % of robot mass) but it costs walking efficiency.
- Peak current densities of 60–95 A/mm² are normal for robot actuators but only for 1.5–4 seconds before the copper heats 40 K (adiabatic estimate). The controller must enforce an I^2t limit, exactly as commercial units do.
- The 12-tooth sun needs a +0.3 profile shift. It is the most heavily loaded gear; contact stress at peak torque is $\approx$ 1406 MPa, so it must be case-carburised and ground (not through-hardened, not printed).

- The JXA-360 driver must handle ≈ 206 A peak. On a 72 V bus (Unitree H1 uses ≈ 67 V) the same motor wound with 1.5 $\times$ the turns needs one third less current, which is the better choice for a full-size robot.

5.2 Prototype shortcut: wind a bought stator core

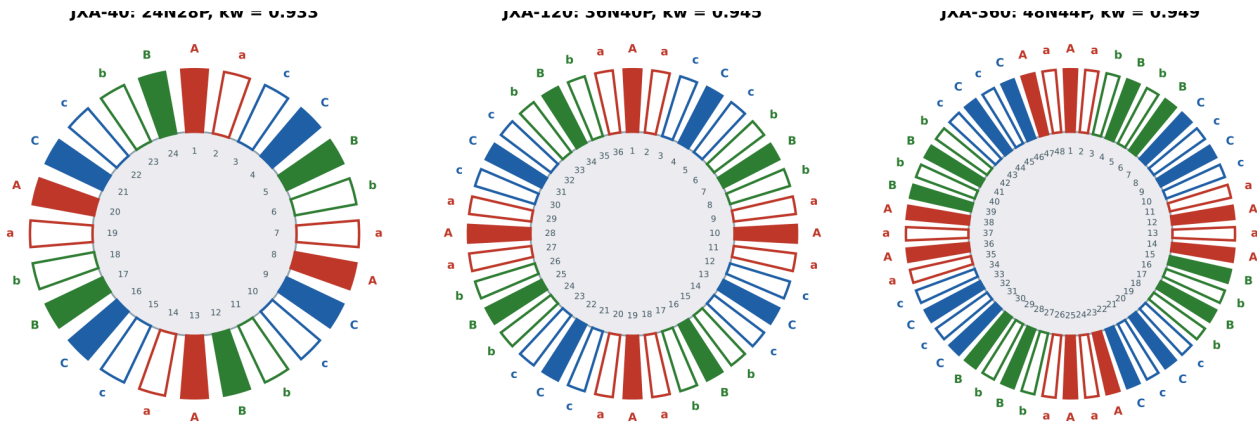
Chinese sellers list bare outer-rotor stator cores of exactly these sizes: "10020" (100 $\times$ 20 mm, 36N42P) for US\$19-31 and "13710" (137 $\times$ 10 mm, 36N42P) for US\$74 (india_motor_materials_raw.md §3.4). Re-running the model with those cores:

Prototype shortcut	Peak	Cont.	No-load	Winding
JXA-120 on a bought 10020 core (100 $\times$ 20 mm, 36N42P, US\$19-31)	124 N·m	44 N·m	23.8 rad/s	14 turns $\times$ 3 $\times$ 0.4 mm, 3 groups, k_w 0.933
JXA-360 on three stacked 13710 cores (137 $\times$ 30 mm, 36N42P, US\$74 each)	372 N·m	147 N·m	18.0 rad/s	8 turns $\times$ 6 $\times$ 0.5 mm, 3 groups, k_w 0.933

For the first prototypes, buying cores removes the hardest tooling step (laminations) and lets you start with winding, which is the actual skill to learn. Check the slot shape on arrival and re-run the script with measured dimensions.

Mass breakdown, g	JXA-40	JXA-120	JXA-360
stator laminations	186	482	1023
copper	45	104	230
magnets	53	125	284
rotor back iron	64	118	285
ring gear	79	179	445
planets	88	233	732
sun and shaft	9	23	55
carrier	69	142	279
housing al	166	273	481
bearings	34	59	108
driver pcb encoders	29	44	70
total g	879	1907	4272

Table 7. Estimated mass by part (g).



Filled tooth / capital letter = wind clockwise (seen from the tooth tip); white tooth / small letter = anticlockwise. Tooth 1 at 12 o'clock, numbering clockwise.

Figure 3. Winding layouts from the star-of-slots calculation, viewed from the tooth tips. Each tooth carries one coil; the letter is the phase, capital = clockwise, small = anticlockwise.

6. Winding the stator: a procedure you can hand to a technician

This is the step people fear most. On a tooth-coil outrunner stator it is slow but simple, and a careful person can learn it on scrap drone stators in one weekend. Any motor-rewinding shop (ceiling-fan, pump or e-rickshaw hub-motor rewinders exist in every Indian city) already knows the skills; give them this page, the layout in Fig. 3 and the table below.

Class	Teeth	Turns / tooth	Wire (bare), strands in hand	Parallel groups	Wire per stator	Copper fill (bare)	R phase @20 °C	Pattern (tooth 1 →)
JXA-40	24	7	0.63 mm × 2 (metric)	1	16 m / 45 g	0.41	69 mΩ	A a c C B b a A C c b B (×2)
JXA-120	36	19	0.60 mm × 1 (23 SWG is 0.610)	4	41 m / 104 g	0.41	48 mΩ	A a c C c b B b a (×4)
JXA-360	48	4	0.85 mm × 3 (metric)	2	45 m / 230 g	0.41	12 mΩ	A a b B b B C c C c a A a A B b B b c C c C A a (×2)

6.1 Tools and materials

- Class-H (200 °C) dual-coat enamelled copper wire in the sizes above (polyesterimide + polyamide-imide, "grade 2" build). Buy a few hundred grams per stator; budget 1.5× the calculated mass for practice and scrap.
- Stator insulation: epoxy powder coating of the lamination stack (0.2–0.3 mm, done by any powder-coating job shop, or a fluidised-bed epoxy kit), or 0.13–0.18 mm Nomex/DMD slot liners cut to shape. Never wind onto bare laminations: the sharp edges cut the enamel.
- A winding jig: the stator clamped on a mandrel in a bench vice or lathe, plus a turn counter. For production, a manual or semi-automatic **needle winder** for outrunner stators (\$6.3 gives prices).
- Wire tensioner (a felt-pad tensioner or simple drag spool), plastic or brass tamping tool (never steel), flat-nose pliers with smooth jaws, PTFE sleeving, Kapton tape, class-H varnish (single-component polyester-imide dip varnish or epoxy trickle resin), an oven that reaches 150 °C, a milliohm meter or 4-wire LCR meter, a 500 V–1 kV insulation tester, and a 0–60 V bench supply.

6.2 Step by step

1. **Prepare the stack.** Deburr the laminations (fine stone, then blow clean). Check the stack is square and tight (bonded, welded or riveted). Powder-coat or fit slot liners. Check with the insulation tester that the coating holds 500 V.
2. **Mark tooth 1** with paint and number the teeth clockwise as in Fig. 3. Write the pattern on a card and tick off each tooth as you go.
3. **Plan the paths.** For one parallel group, phase A's teeth are wound one after the other with one continuous wire, following the pattern: capital letter = wind clockwise (looking at the tooth tip), small letter = anticlockwise. Going from an "A" tooth to the next "a" tooth you simply keep winding in the opposite sense; do not cut the wire. Leave 10 cm tails.
4. **Wind the first layer tight against the tooth**, turns side by side, starting at the slot bottom (near the yoke) and working outwards. Keep steady tension (roughly 10–15 % of the wire's breaking load: about 15 N for 0.6 mm wire). Lay the crossover between teeth along the end face, never across the air-gap face.
5. **Count turns exactly.** One turn too many or too few on one tooth makes the phases unequal: current ripple, noise and heat. Use a counter on the jig.
6. **Keep half the slot for the neighbour.** Each slot holds two coil sides (one from each tooth). Stop each coil at the slot's centreline; tamp the turns with the plastic tool. Our fill target is 0.42 bare-

copper fill; if the last turns do not fit, the wire is too thick: change to more strands of thinner wire with the same total copper area.

7. **Wind B and C the same way**, starting 120 electrical degrees later as the pattern shows. With several parallel groups, wind each group (one repeating unit of the pattern) as a separate continuous wire.
8. **Terminate**. Scrape or burn off the enamel at the tails (solder pot at 450 °C or a stripping cream), twist the strands, solder or crimp. Join the three "end" tails into the star point, sleeved in PTFE. Join parallel groups at the phase terminals. Tie the end turns with polyester lacing cord.
9. **Test before varnishing** (fix mistakes now, it is impossible later):
 - phase-to-phase resistance: all three within ± 2 % of each other and within ± 10 % of the table;
 - insulation: >100 M Ω at 500 V DC from every phase to the stack;
 - inductance with an LCR meter at 1 kHz: equal within ± 5 % (a wrong-direction coil shows up here);
 - after the rotor is fitted: spin it by hand (or with a drill) and look at the three line-line back-EMFs on an oscilloscope: three equal, clean sine waves 120° apart. Their amplitude gives K_e directly.
10. **Varnish (impregnate)**. Pre-heat to 100 °C, dip or trickle class-H varnish until it stops bubbling, drain, then cure (typically 2 h at 150 °C; follow the varnish datasheet). Varnish locks the turns (no rubbing, no noise), fills the air pockets and roughly halves the winding-to-iron thermal resistance.
11. **Re-test** (resistance, insulation, back-EMF) and record the numbers with the stator's serial number.

6.3 Do it yourself, give it to a rewinder, or use a machine?

Route	Time per JXA-120 stator	Cost	Quality risk	When
Hand winding on a jig (you)	3-5 h (the first ones 8 h+)	labour only	turn-count errors, low fill	prototypes, 1-25 units
Local motor rewinder (job work)	1-2 days turnaround	₹800-2,000 per stator (ESTIMATED)	wire grade swaps, poor insulation: inspect and test	batches of 5-50
Manual/semi-auto needle winder	20-40 min	machine ₹2-7 lakh (US\$5-8k semi-auto)	low once set up	50-500 / year
CNC multi-spindle needle winder (China)	1-3 min	machine ₹15-30 lakh (US\$15-30k)	low	> 5,000 / year

Practise first: buy two or three cheap 5010/8108-class drone stators, unwind them (count the turns), and rewind them. When your rewound motor's resistance and back-EMF match the original, you are ready for the real stator.

7. Making every other part: the in-house route, step by step

"In-house" here means: design, winding, assembly, magnetising, calibration and testing under your roof, with precision cutting (laminations, gears, housings) bought as **job work** from Indian shops. Nobody should buy a gear-hobbing machine or a lamination press for 25 actuators.

7.1 Parts list and who makes each part

Part	Material / spec	How to make (prototype)	How to make (volume)
Stator laminations	CRNO M270-35A / 35C250 0.35 mm (0.2 mm for lower iron loss)	fibre-laser or wire-EDM cut stack, then bond with lamination varnish or weld 3 seams; anneal if possible	progressive stamping die + interlocking
Magnets	NdFeB N42SH/N45SH arc or flat segments, Ni-Cu-Ni plated, magnetised radially	buy pre-magnetised segments (China or Indian distributor)	same, by the thousand

Rotor can	EN3/EN8 (1018/1045) mild steel tube or bar	CNC lathe job work	CNC lathe / deep-drawn cup
Sun, planets	EN353 or 20MnCr5, carburised 58–62 HRC case	gear-hobbing job shop (+ heat treat) or wire-EDM from pre-hardened blanks	hobbing + grinding, or powder metallurgy
Ring gear	EN19/42CrMo4 nitrided, or EN353 carburised	wire-EDM the internal teeth (easiest), or shaping	broaching / power skiving
Carrier, housing, hub	Al 6061-T6 / 7075-T6 (or printed AlSi10Mg)	CNC milling/turning job work	sand casting (≥ 20 pcs) or die casting (≥ 5 –10k/yr) + finish machining (\$7.4)
Output bearing	crossed-roller (CRBH/RU) or two thin-section 68xx/69xx	buy	buy
Encoders	MT6835/MA600 on the rotor, MT6701/AS5047P on the output, diametric magnets	on the driver PCB	same
Driver	STM32G4 + 3-phase gate driver + 6 $\times$ 80–100 V MOSFETs + shunts + CAN transceiver	PCB + assembly at Lion Circuits / JLCPCB	same, panelised

7.2 The build sequence

1. **Laminations.** Send the DXF of one lamination to a fibre-laser shop that cuts electrical steel. Ask for nitrogen-assist cutting and no burr. Cut 3–5 % extra. Stack on a mandrel with a key, press flat, bond with lamination varnish (or weld 3 shallow seams along the back of the yoke), then powder-coat. The laser's heat damages the steel near the cut and raises iron loss by roughly 10–30 % on narrow teeth; acceptable for prototypes. Wire-EDM through a clamped stack avoids this but costs more per stack.
2. **Wind and varnish** the stator (Section 6).
3. **Rotor.** Clean the rotor can and magnets with isopropyl alcohol. Bond the magnets with high-temperature structural epoxy or anaerobic magnet adhesive (e.g. Loctite AA 3342 / Araldite 2014 class), alternating N and S, using a non-magnetic (printed or aluminium) spacer comb so they stay equally spaced. Check polarity with a compass or a gauss meter as you go; one reversed magnet ruins the motor. Optionally wrap with glass-fibre/epoxy or Kevlar for retention. Balance the rotor on a simple knife-edge or by drilling the end plate.
4. **Gears.** Order sun, planets and ring from a gear job shop with the tooth data from Table 6 (module, teeth, pressure angle 20°, profile shift +0.3 on the 12-tooth sun, quality DIN 7 or better, case-carburised 0.4–0.6 mm). Check the planet bores and pins for a sliding H7/g6 fit; use needle bearings or bronze bushes in the planets.
5. **Housings.** CNC job work in 6061-T6, anodised (from ≈ 20 pieces, a sand-cast LM25 blank finish-machined is cheaper: \$7.4). Tolerances that matter: bearing seats (± 0.01 mm), ring-gear seat concentric to the output bearing seat (≤ 0.02 mm), and the stator hub concentric to the rotor bearings (the air gap is only 0.55 mm).
6. **Driver PCB.** Two encoders, the FOC microcontroller and the power stage go on one round board on the back of the actuator. Reuse an open design for the first version (B-G431B-ESC1 class, moteus, or the MIT Mini Cheetah/Ben Katz driver); keep the same CAN protocol as RobStride (MIT mode) so JX1's software does not change.
7. **Assemble.** Fit the stator to the hub; press the rotor bearings; insert the rotor *with a guide sleeve*: the magnets pull the rotor sideways with hundreds of newtons and will snap onto the stator and chip if you let go. Fit the gear stage with grease (lithium-complex or synthetic EP grease), the carrier and the output bearing, then the driver.
8. **Calibrate.** Encoder offset and linearisation, phase order, current-loop tuning, cogging-torque compensation table, output-encoder offset for absolute joint position.

9. **Test on a bench** (Section 9): back-EMF, K_t with a torque arm and a load cell, peak torque, 1 h thermal soak at continuous torque, backlash, then a 50-hour gait-trace life test on at least one unit per class.

7.3 Safety

- NdFeB magnets pinch and shatter. Wear eye protection; keep them away from phones, pacemakers and watches; store them in keepers. Machine or grind them only wet: the dust burns.
- A 48 V battery can deliver hundreds of amperes into a short. Fuse every actuator, and test new drivers through a current-limited supply first.
- Varnish and epoxy fumes: work in ventilation, and use the oven only for curing, never for food.

7.4 Casting and moulding: which process for which part

"Moulding" and "casting" mean making a part by filling a hollow shape (a mould) with liquid material that then hardens. The process you choose depends on the material, how many parts you need and how precise they must be. Data: research/raw/india_casting_moulding_raw.md (IndiaMART/TradeIndia listings, vendor pages, 6 Oct 2026; ESTIMATED unless quoted).

What does not work, and why

- **Molten aluminium into a silicone mould.** Aluminium melts at 660 °C and is poured at 700–750 °C; the best heat-resistant silicones survive ≈ 300 °C (Mold Max 60 ≈ 294 °C). The rubber burns and gasses. Silicone moulds are for resin (urethane casting, ≈ 20 –25 parts per mould) or low-melting metals (pewter ≈ 170 –230 °C, Field's metal ≈ 62 °C), which are too soft for an actuator.
- **Molten metal into cement or concrete.** Cement always holds water; water touching molten metal flashes to steam $\approx 1,600\times$ its volume and throws metal metres away. Never pour into cement, damp sand, wet scrap or onto a bare concrete floor.
- **"Injection moulding" of metal housings.** Injection moulding pushes hot *plastic* into a steel mould (₹1.5–15 lakh for a part this size). A glass-filled nylon or PPS housing costs only ₹50–280 a part, but plastic cannot hold a ± 0.01 mm bearing seat or carry motor heat, so it needs metal inserts and only makes sense for covers or for very large volumes.

The processes that do work

Process	How it works (plain words)	One-time cost	Per part (raw)	Tolerance	Use it for
Sand casting	a pattern (3D-printed PLA/PETG is fine) is pressed into packed sand; molten aluminium (LM25/LM6) fills the cavity	pattern ₹100–5,000 (lasts 10–500 moulds)	₹250–420/kg → ₹150–380 per housing	± 0.5 –1 mm, rough surface	housings from ≈ 20 pieces
Investment (lost-wax / lost-PLA)	a wax or PLA copy is covered in a plaster-silica shell, burnt out in a kiln, then metal is poured in the hollow	wax die ₹10–45k (or none with printed PLA)	Al $\approx$ ₹350/kg; steel ₹250–650/kg	± 0.2 –0.5 mm, fine detail	complex housings, brackets, small batches
Gravity die casting	molten metal poured into a reusable steel mould	₹0.45–2 lakh	₹310–600/kg	± 0.3 mm	500–5,000 / yr
High-pressure die casting	molten aluminium injected at high pressure into a hardened steel die	₹2–6 lakh	₹200–450/kg	± 0.14 mm; no T6 heat treat	≥ 5 –10k / yr
Urethane / vacuum casting	liquid resin poured into a silicone mould made from a printed master	₹1.6–7.6k of silicone per mould	₹1,000–5,000 per part (service)	± 0.2 mm	plastic covers, 10–25 parts
Plastic injection moulding	hot plastic injected into a steel/aluminium mould	₹1.5–15 lakh	₹50–280 (PA66-GF30 / PPS)	± 0.05 –0.1 mm	covers, cable parts at ≥ 5 k / yr

Metal injection moulding (MIM)	metal powder + binder moulded, then sintered	mould ₹2.3–2.8 lakh	₹20–150	±0.3–0.5 %	small parts < 50 g, e.g. planets, at ≥ 5k / yr
Powder metallurgy (sintered) gears	steel powder pressed in a die and sintered (7.0–7.3 g/cm ³)	die (no Indian quote found)	₹5–100 per gear	DIN 8–9	planets/sun of light-duty classes at ≥ 10k / yr

JXA-120 housing route (finished, ESTIMATED)	2 pcs	25 pcs	200 pcs	1,000 pcs	10,000 pcs
CNC from billet (6061-T6) Ø125 × 75 bar ≈ 2.5 kg for a 350 g part	₹5,960	₹2,004	₹1,242	₹1,034	₹1,029
Sand casting (LM25) + finish machining printed pattern ₹5k (100–500 moulds) + fixture ₹8k; casting ₹250–420/kg	₹8,426	₹1,585	₹732	₹548	₹536
Gravity die casting + finish machining steel die ₹0.45–2 lakh; ₹310–600/kg	₹65,895	₹6,238	₹1,399	₹767	₹652
High-pressure die casting + finish machining die ₹2–6 lakh, pays off at 5–10k pcs; no T6	₹2,05,350	₹17,072	₹2,516	₹792	₹425
Metal 3D printing (AlSi10Mg) + finish machining ₹15–30/g, ≈ 455 g printed incl. supports/stock	₹12,620	₹10,762	₹10,436	₹10,350	₹10,348

Table 7b. Cost per finished JXA-120 housing by route and lot size (cost_model.py; cheapest in bold). Machining rate falls from ₹2,500/h for tiny lots to ₹400/h at volume.

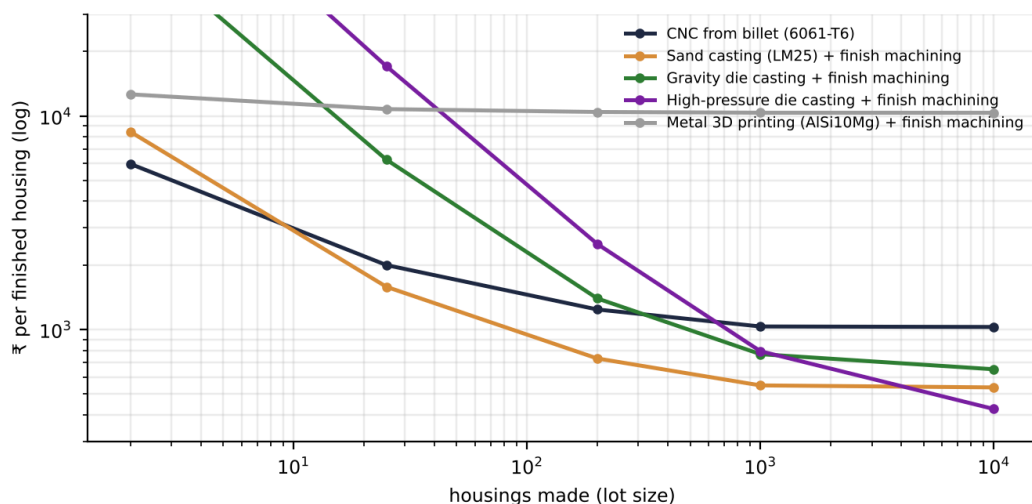


Figure 3b. Finished-housing cost vs lot size. Billet CNC wins below ≈ 10 pieces, sand casting from ≈ 20, high-pressure die casting only near 10,000.

What this means for the JXA actuators

- **Housings:** machine from billet for the first 5–10 (the design will change). From ≈ 20–25 pieces, a **sand-cast LM25 housing from a 3D-printed pattern, then finish-machined**, is the cheapest route; at 10,000 a year high-pressure die casting wins. Always leave 1–2 mm of machining stock on the bearing seats, the ring-gear bore and the mounting faces, and ask the foundry for vacuum impregnation if the housing must be sealed.
- **The saving is small:** ≈ ₹400–500 per housing at 25–200 pieces, about 1.5–2 % of a batch-built JXA-120. Casting cuts the waste metal ≈ 6×, but machine time only ≈ 25–30 %, because the bearing seats and the two set-ups dominate either way. Do it for weight and shape freedom (fins, bosses, ribs), not as the main cost lever.
- **Gears are the bigger lever at volume:** a hobbed, carburised gear set costs ₹1,800–5,500, while sintered or MIM planets cost ₹20–150 each once tooling is paid (≥ 5–10k / yr). Sintered steel has lower fatigue strength than wrought steel, so keep the 12-tooth sun (≈ 1,400 MPa contact stress)

hobbed and case-hardened, and try sintered planets first on JXA-40. RobStride's budget EduLite series already uses sintered gears.

- **Never** cast gears, shafts or the stator.

If you want to cast aluminium yourself (lost-PLA)

1. Print the housing in PLA with 1-2 mm extra stock on machined faces; add a sprue (pouring channel) and vents.
2. Invest it in a proper casting investment (Plasticast / Ultravest / SRS class, ≈ ₹3,200 per 22.5 kg bag), not plain plaster of Paris, which cracks.
3. Burn out in a programmable kiln (₹24.5-55k): slowly to ≈ 150 °C, then 400-730 °C over 5-12 hours, until no PLA ash is left.
4. Melt clean aluminium (LM25 ingot, not random scrap) in a crucible furnace (≈ ₹18k for 5 kg, 1,200 °C), degas, pour into the **hot** mould.
5. Break out, cut the sprue, then send it for finish machining.

Safety first: face shield, leather or aluminised apron and gloves, leather boots, cotton clothes (no synthetics), a dry sand floor, preheated tools and moulds, and a respirator when mixing silica investment. Start with a foundry; learn to cast yourself only after the design is frozen.

8. What it costs: make in India vs buy from China

All prices are ESTIMATED from listings read on 24 Sep-6 Oct 2026 (sources in the raw logs) at ₹95.96/US\$ and ₹13.52/CNY; imports carry the repository's landed factor (duty 32.3 % × freight 7 %). One aggregator quotes a higher motor duty (≈ 40 %); confirm on ICEGATE before ordering.

8.1 How Chinese makers get so cheap

- **Volume.** RobStride (founded Nov 2023 by Xiaomi's former CyberGear team) passed 50,000 joints shipped by Sep 2025 and guides 300,000 units for 2026 at an average ≈ ₹1,000 each. Encos shipped > 100,000 in 2025; Quanzhibo runs a 90-second-per-joint, > 85 %-automated line.
- **Bill of materials, not wages.** Broker teardowns put a Unitree G1 joint at ₹1,000 (small) to ₹1,500 (large): reducer ≈ 27 %, motor ≈ 20 %, encoders ≈ 20 %, driver ≈ 17 %, structure ≈ 16 %. Direct labour is only 8-14 % of Unitree's cost; Chinese factory wages (≈ ₹86k/month) are 3.5-4.5× India's. **India's lower wages cannot close a gap that comes from volume and a dense local supply chain.**
- **Everything is next door.** Stamped laminations, sintered magnets (China makes ≈ 94 % of the world's), hobbed gears, crossed rollers (Luoyang), die-cast housings and PCB assembly are all bought within a day's drive, at volume prices. In India, magnets and most crossed-roller bearings are 100 % imported; gear and lamination job shops exist but quote one-off prices.
- **Magnets are cheap; access is the risk.** At bulk prices the magnets in a JXA-120 cost ≈ ₹750. China's April 2025 licence rules for Dy/Tb (high-temperature SH grades) still apply; the wider October 2025 rules are suspended only until **10 November 2026**. India's ₹7,280 crore magnet scheme (approved Nov 2025) has bids but no plant yet; expect no Indian supply before 2027-28.

Our estimate of a Chinese maker's manufacturing cost (COGS) for RS04 is ¥660-1,500, central ¥840 (≈ ₹11,357), from the list price minus a 30-45 % gross margin, cross-checked with the G1 teardowns.

8.2 Cost per actuator

Class	Make: prototype	Make: batch 25/yr	Make: 1,000/yr	Buy: USD list, landed	Buy: China distributor, landed	China maker's cost (COGS, est.)
JXA-40 RobStride RS06 (36 N·m)	₹52,845	₹19,777	₹7,479	₹28,522	₹17,059	₹6,354-₹13,520

JXA-120 RobStride RS04 (120 N·m)	₹77,353	₹30,292	₹11,493	₹34,634	₹24,091	₹8,923–₹20,280
JXA-360 Damiao DM- J10422P (400 N·m)	₹1,46,564	₹61,822	₹23,322	₹65,194	–	₹20,280–₹40,560

Table 8. Direct cost per actuator (parts + labour + 5–15 % test/scrap), vs buying. The batch tier assumes job-work parts and a technician at ₹300/h; the factory tier assumes stamped laminations, bulk magnets/cores and a semi-automatic line.

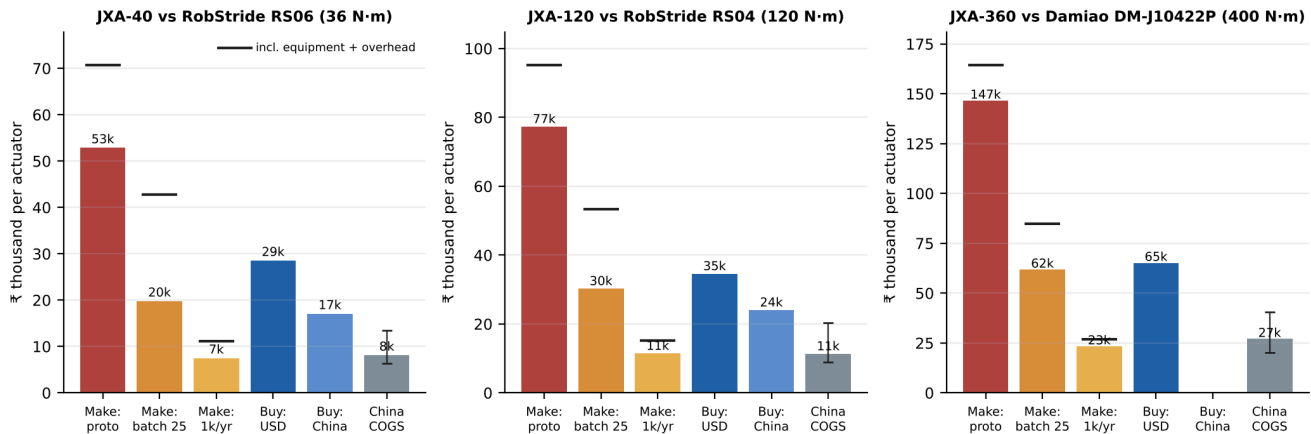


Figure 4. Per-unit cost: making (red/orange/yellow) vs buying (blue) vs what it costs a Chinese maker (grey, with range). Black bars add equipment (3-year write-off) and overhead.

Fully loaded (equipment over 3 yr + overhead)	Prototype	Batch	1,000/yr	₹/N·m proto	₹/N·m batch	₹/N·m 1k/yr
JXA-40	₹70,734	₹42,764	₹11,153	₹1710	₹1034	₹270
JXA-120	₹95,241	₹53,279	₹15,167	₹762	₹426	₹121
JXA-360	₹1,64,452	₹84,808	₹26,997	₹443	₹229	₹73

Table 9. Fully loaded cost per unit and per N·m of peak torque.

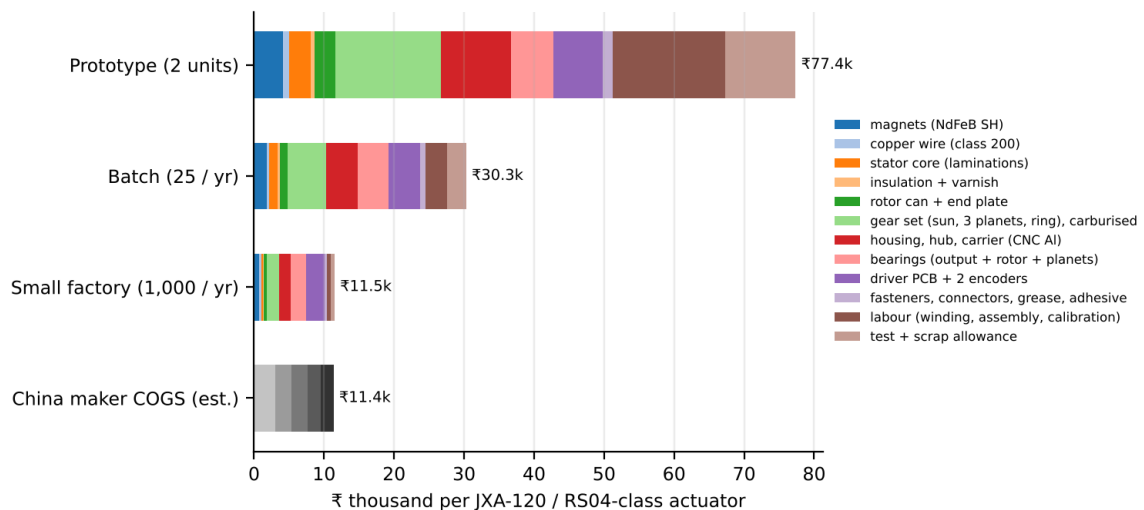


Figure 5. Where the money goes in a JXA-120. Prototypes are dominated by gears, machining and your own hours; at 1,000/yr the bill looks like the Chinese one.

JXA-120 line item	Prototype	Batch 25/yr	1,000/yr
magnets (NdFeB SH)	₹4,128	₹1,876	₹751
copper wire (class 200)	₹935	₹312	₹265
stator core (laminations)	₹3,000	₹1,200	₹300

insulation + varnish	₹600	₹250	₹80
rotor can + end plate	₹3,000	₹1,200	₹400
gear set (sun, 3 planets, ring), carburised	₹15,000	₹5,500	₹1,800
housing, hub, carrier (CNC Al)	₹10,000	₹4,500	₹1,600
bearings (output + rotor + planets)	₹6,100	₹4,400	₹2,250
driver PCB + 2 encoders	₹7,000	₹4,500	₹2,600
fasteners, connectors, grease, adhesive	₹1,500	₹800	₹350
labour (winding, assembly, calibration)	₹16,000	₹3,000	₹550
test + scrap allowance	₹10,089	₹2,754	₹547
Total	₹77,353	₹30,292	₹11,493

Table 10. JXA-120 line items.

8.3 Equipment you need (one-time)

You do not need a factory to start. The garage kit below makes and tests prototypes; the workshop add-ons take you to batches; only the third list is a real factory, and it only pays at $\approx 1,000$ units per year (Section 8.6).

A. Garage / prototype kit: ₹3.2 lakh

Item	₹
winding jig + turn counter	₹5,000
LCR meter (TH2830 class)	₹45,000
insulation (hipot) tester	₹15,000
4-wire milliohm meter	₹10,000
bench supply 60 V / 15 A	₹18,000
curing oven (used)	₹15,000
oscilloscope	₹35,000
dyno bench: hysteresis brake + DYN-200 torque sensor + frame	₹1,40,000
USB-CAN adapters	₹6,000
printed assembly fixtures, rotor guide sleeves	₹8,000
hand tools, consumables	₹25,000

B. Workshop add-ons (batch): ₹12.5 lakh

Item	₹
semi-automatic needle winder (US\$5-8k + duty)	₹7,00,000
surge (impulse) tester	₹1,00,000
second dyno + thermal chamber	₹1,50,000
assembly + magnet-bonding fixtures	₹1,50,000
simple rotor balancer	₹50,000
inspection: bore gauges, height gauge, gear-roll tester	₹1,00,000

C. Small factory add-ons (1,000+/yr per class): ₹1.35 crore

Item	₹
CNC needle winders x2	₹40,00,000
stator stamping dies, 3 classes (no quote found)	₹24,00,000
gear tooling (hobs, PM dies for sun/planets)	₹15,00,000
end-of-line test stations x2	₹20,00,000
dynamic balancing machine	₹6,00,000
varnish/cure line	₹5,00,000
ESD assembly line + fixtures	₹15,00,000
quality lab	₹10,00,000

8.4 Metal 3D printing

Indian DMLS services (iamRapid, Bengaluru: AlSi10Mg $\approx$ ₹15–30 per gram, i.e. ₹40–80/cm³, 7–10 working days; Wipro3D and others on quote) can print the housing, hub and carrier. For a JXA-120 that is ≈ 150 cm³ of AlSi10Mg, roughly ₹6,000–12,000 plus post-machining of the bearing seats and the ring-gear bore (printing holds ± 0.1 mm; bearing seats need ± 0.01 mm). That is **no cheaper than CNC from billet** and slower, so print only shapes CNC cannot make (lattice-lightened housings, integrated cooling fins). **Do not**

print gears: printed steel (316L $\approx$ ₹160/cm³ at JLC3DP) has neither the case hardness nor the surface finish a carburised sun gear needs at 1,400 MPa contact stress.

8.5 Scenarios

JX1: legs + waist (13 actuators)	₹
Buy: 4 × RS04 + 2 × RS03 + 7 × RS06, USD list landed (current BOM)	₹3,99,310
Buy: same via China distributor, landed	₹2,55,040
Make: 7 × JXA-120 + 8 × JXA-40 (incl. 1 spare each) at batch cost	₹3,70,258
+ 2 prototype rounds × 2 units × 2 classes	₹5,20,789
+ garage equipment kit	₹3,22,000
Make: first set, all-in	₹12,13,048
Make: every further set (13 actuators, batch cost)	₹3,20,190

Full-size humanoid (28 actuators: 4 × 360, 14 × 120, 10 × 40 N·m)	₹
Buy: Damiao J10422P + RS04 + RS06, USD list landed	₹10,30,884
Buy: same, RobStride via China distributor	₹7,68,647
Make: prototype prices	₹21,97,637
Make: batch prices (direct) / fully loaded	₹8,69,142 / ₹15,12,769
Make: 1,000/yr prices (direct) / fully loaded	₹3,28,974 / ₹4,31,864
China maker's cost for the same set (COGS, est.)	₹3,48,275

8.6 Break-even

Against an RS04 bought through a China distributor (₹24,091 landed):

- **Batch tier:** each in-house JXA-120 costs ₹30,292 in parts and labour, **₹6,200 more than buying** before any equipment or overhead. There is no break-even at this scale: making 25 actuators a year is a learning and independence investment, not a saving.
- **Factory tier:** at 1,000/yr the direct cost falls to ₹11,493, ₹12,599 below the landed RS04 price. With ₹1.10 crore of yearly fixed cost (equipment written off over 3 years + engineers, QA, rent), break-even is **$\approx$ 875 JXA-120-class actuators a year** (fewer if the fixed cost is shared with the other classes). Above that, an Indian maker beats the *landed* import, but still not RobStride's own cost ($\approx$ ₹11,357), so it would need to compete on duty, service, lead time and supply security, not on price.
- **Where in-house wins clearly:** the 360 N·m class. Nobody sells it cheaply (RobStride has none; Damiao J10422P lands at ₹65,194), and an in-house batch unit (₹61,822) is already close to that.

9. Testing: how you know your actuator is good

Test	How (garage kit)	Pass criterion
Winding resistance, balance	4-wire milliohm meter, each phase pair	within ± 2 % of each other, ± 10 % of design
Insulation	500 V–1 kV insulation tester, phases to stack	> 100 M Ω ; no breakdown at $2 \times V_{dc} + 1,000$ V hipot (production)
Inductance balance	LCR meter at 1 kHz	± 5 %
Back-EMF / K_e	spin with a drill, scope the line voltages	three equal sines, $K_e \pm 10$ % of design, harmonic distortion < 5 %

Cogging torque	torque sensor, rotate slowly by hand	< 1 % of rated torque at the output (after compensation)
K_t and peak torque	lever arm + load cell or DYN-200 sensor; locked output; current steps up to peak	torque vs current straight to $\approx I_{pk}/3$, then within 15 % of design at peak
Torque-speed	hysteresis brake load at 48 V	matches Fig. 2 within ± 10 %
Thermal	continuous torque on an aluminium plate, 1 h, thermistor in winding	winding stabilises < 120 °C
Backlash, stiffness	lock motor, dial gauge on an arm, ± 10 % rated torque	< 15 arcmin backlash (RobStride class: 6-18)
Life	replay JX1 walking torque traces on the dyno (57 h is the longest published DIY test)	≥ 200 h at 50-80 % rated torque with no backlash growth > 50 %

10. Risks

Risk	Effect	Mitigation
Chinese rare-earth export rules (SH grades with Dy/Tb need a licence; wider rules return after 10 Nov 2026 unless extended)	no high-temperature magnets, months of delay	buy the first 200-300 magnet sets early; design with N48H (no Dy, needs a lower temperature limit, ≤ 100 °C); watch the Indian REPM scheme plants
Gear quality from local job shops	noise, backlash, early pitting of the 12-tooth sun	specify DIN 7, profile shift, case depth and hardness on the drawing; inspect with a gear-roll tester; first lot from two shops
Winding errors	unequal phases, hot spots, failed hipot	test before varnish (Section 6.2 step 9); jig with turn counter
Heavier than RobStride	more inertia in the legs, less battery life	second iteration: compound planets, lightened carrier, die-cast housing, 0.2 mm laminations
Driver development time	months of firmware work	start from an open driver (moteus / Ben Katz / B-G431B-ESC1 class) and keep RobStride's MIT-mode CAN frames
Hidden cost of your time	a year of engineering is the biggest real cost	run the program in parallel with buying JX1's actuators; do not put JX1 on the critical path of a new actuator

11. Plan: from a rewind drone motor to your own actuator line

Phase	What	Time	Cash (ESTIMATED)	Exit test
0. Learn	Buy 2-3 cheap 8108/10010-class stators and one RS04 (to measure and take apart). Rewind a stator; drive it with an open FOC board.	1 month	₹60k-1 lakh	your rewind motor matches its original R and K_e within 5 %
1. Garage kit + JXA-120 v1	Buy the test kit; bought 10020 core, bought magnets, job-shop gears and housings; 2 units.	3-4 months	₹4.8 lakh	Table in Section 9 passes; 50 h gait-trace test
2. JXA-40 v1 + JXA-120 v2	Own laser-cut laminations; lighter planets; own driver PCB on the back.	3 months	₹2.6 lakh	mass ≤ 1.6 kg (JXA-120); 200 h life test
3. One leg	Fit JX1's knee and hip pitch with JXA-120 (the Modular Actuator Interface keeps the bolt pattern).	2 months	₹1.2 lakh	leg passes JX1's bench walking test
4. Batch	Workshop add-ons; 15 actuators for a full JX1 set.	3 months	₹16.2 lakh	robot walks on in-house actuators
5. JXA-360	Full-size knee class for the next robot.	4 months	₹5.9 lakh	360 N·m, 15 rad/s on the dyno

12. Conclusions and recommendation

1. **You can build a RobStride-class actuator yourself.** The physics is the Lorentz force and a careful slot/pole/turns calculation; our analytic JXA-120 lands within 3–20 % of RS04's published constants. The only skilled manual job is winding, which a careful person or any motor-rewinding shop can do with the procedure in Section 6.
2. **You do not need a factory.** A ₹3.2 lakh garage kit, bought stator cores and magnets, and Indian job shops for gears and housings are enough for prototypes.
3. **It will not be cheaper at small volume.** A prototype JXA-120 costs $\approx$ ₹77,353 and a batch unit $\approx$ ₹30,292, against ₹24,091 for a landed RS04 and $\approx$ ₹11,357 for RobStride's own cost. An Indian line only beats the import at about 875 actuators a year, and it never beats the Chinese maker's cost unless volume reaches Chinese levels.
4. **Recommendation:** keep buying RobStride for JX1 now (the robot should not wait for a new actuator); start the in-house program in parallel at Phase 0–1 as the "V2 localisation" path already planned in `actuators/actuator_selection.md`. Aim it first at the **360 N·m class**, where nothing cheap exists and the in-house cost is already near the import price, and at **supply independence** from rare-earth and export-rule shocks.

References and data

Every number in this paper is labelled in the underlying logs: **VERIFIED** (official page or primary document), **CALCULATED** (our model), **ESTIMATED** (our arithmetic on cited prices), **UNVERIFIED** (secondary source).

- Design model and figures: `actuators/inhouse/qdd_design.py` → `results/design.json`, `results/variants.json`
- Cost model: `actuators/inhouse/cost_model.py` → `results/cost.json`
- Actuator survey ($\approx$ 110 units, prices, specs): `research/raw/actuator_technology_raw.md`
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- India / China material, part, equipment prices: `research/raw/india_motor_materials_raw.md`, `india_mechanical_manufacturing_raw.md`, `india_electronics_compute_raw.md`
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